BenchMatch / Round 1 proposal

Open Innovation

Authored candidate - official validation and NAS execution pending

1. What does the Skill do?
BenchMatch compares documented conditions of two prototype runs and preserves differences, missing values and source conflicts in a reviewed report.

2. Target user and problem
Makers and small workshops keep run notes in scattered files and often repeat a failed test without knowing what changed.

3. Proposed solution
Reconcile run-bound parameters and reported outcomes, expose gaps and prepare a consistent recording checklist for the next test.

4. Core user flow
Identify comparable runs and variables; authorize sources; inspect differences and unknowns; add or clarify a note; approve the report.

5. Substantive difference
The workflow preserves variable-level uncertainty and run identity instead of turning a plausible explanation into an unsupported cause.

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6. UGREEN AI NAS capabilities

File discovery establishes run sources; document extraction and reconciliation recover values and units; optional supported transcription adds a note; approved text saving preserves the comparison.

7. Technical approach

A Markdown parent delegates to file, document and knowledge experts with actual source results. Inputs include run identities, shared context, variables, units and explicit tolerances. CLI bindings need official-kit verification. Confirm model routing before private reads; prefer verified local processing. No equipment control, external API or custom runtime is bundled. Codex assisted development.

8. Current development status

Implemented: workflow, comparison rules, references, synthetic run records, HTML concept demo, proposal and local package checks. Official-tool use and validation remain blocked by download access. NAS execution, extraction, approvals and model routing are untested; entrant review is pending.

9. Final validation plan

Core one: identify temperature and lot changes while keeping missing humidity unknown. Core two: incorporate a run-linked humidity note and retain conflicting temperatures. Edge: incompatible units or unresolved run identity must prevent an unjustified comparison.

10. Known limitations

Incomplete records and extraction errors limit findings. Comparable differences do not establish causality. Unit mappings require explicit support, and the workflow does not prescribe machine settings or certify an experiment.

Third-party information (excluded from substantive response limit)

UGREEN tools/experts: UGREEN terms; no kit redistributed. No third-party runtime code, dataset or model bundled. Separate demos: Rubik, SIL OFL 1.1. PDF creation: original Python standard-library code. Local checks: pypdf, BSD-3-Clause; PyYAML, MIT. Sources and full disclosures accompany the Skill.